Differentiation Practice

For 1–21 below, find $\frac{dy}{dx}$. For 22–25 below, find $\frac{d^2y}{dx^2}$. For each question, also circle all the differentiation rules you used. In all cases, show all your work.

1. $y = x^3 - \frac{5}{x} + \pi$

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Exponential/log derivative(s)

2. $y = \sqrt{2x} + \sqrt[3]{x+1}$

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3. $y = (x^2 + 4x)^{\frac{3}{2}}$

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4. $y = (x-1)^5(x^2+2)^{13}$

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5. $y = \frac{x^3-x}{x^2-9}$

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6. $y = (4x + \frac{1}{x^2})^3$

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7. $y = x^3(x^2+1)\sqrt{x-1}$

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8. $y = (x^2+1)(3x+1)$

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9. $y = \sqrt{\frac{1}{2}x^3}$

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10. $y = (x^2 + 1)(x + 1)^2$

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11. $y = (\sqrt[3]{x} - x)^{\frac{4}{3}}$

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12. $y = \sqrt[5]{3x^4 + x^2}$

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13. $y = (x^{78} - 77x)^{87} - x^{87}$

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14. $y = \left(\frac{x^2+2}{3}\right)^2$

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15. $y = \sqrt{\frac{x^2+9}{x+3}}$

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16. $y = \frac{(\sqrt{x}-1)(x+1)}{3x^2}$

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17. $y = 3^{-\pi^2}$

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18. $y = (x^2 - x)^{-2}$

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19. $y = \frac{x+\sqrt{x}}{x^2+\pi}$

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20. $y = (x^2 - x)\sqrt{5x+1}$

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21. $y = \frac{1}{x^3+2x}$

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For the next four, find $\frac{d^2y}{dx^2}$ (aka y'', aka $f''(x)$):

22. $y = (x^2 + 1)(x^2 - 1)$

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23. $y = 4^{\pi}(x^2 + \frac{1}{x})$

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24. $y = \frac{\sqrt{x}}{1-x}$

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(Hint: after you find the first derivative, multiply numerator and denominator by $2\sqrt{x}$ and simplify before you find the second derivative.)

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Exponential/log derivative(s)

25. $y = (\sqrt{x} - 1)^{100}$

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